R Practice Questions: Missing Values (NA) 50mks

Q1. Using `is.na()` and `colSums()`, count the total number of missing values in the entire dataset and display the NA count per column in descending order.

Q2. Calculate the percentage of missing values for each column and display only columns that have more than 10% missing data.

Q3. Count the number of rows that contain at least one missing value and the number of rows that are completely complete with no NAs.

Q4. Create a custom summary table using `sapply()` that shows each column's name, data type, total values, non-missing count, and missing count.

Q5. Using `barplot()`, create a bar chart showing the number of missing values per column, with each bar labelled by its exact NA count.

Q6. Filter the dataset to return only rows where both `Age` and `Cholesterol` are not missing, and state how many rows remain.

Q7. Filter the dataset to return rows where any of `BloodPressure`, `BMI`, or `Glucose` contain a missing value, and display the `PatientID` for those rows.

Q8. Using `dplyr::filter()`, return only rows where `Diagnosis` is `"Diabetes"` and `Glucose` is not missing, then state how many such patients exist.

Q9. Replace missing values in the `Age` column with the mean age (rounded to 1 decimal) and replace missing values in `Gender` with the string `"Unknown"`, then confirm the changes using `summary()`.

Q10. Using `dplyr::mutate()` and `replace_na()`, replace missing `Cholesterol` with its median, missing `BMI` with its mean, missing `HospitalDays` with `0`, and missing `SmokingStatus` with `"Not Reported"`, then verify zero NAs remain in those columns.